

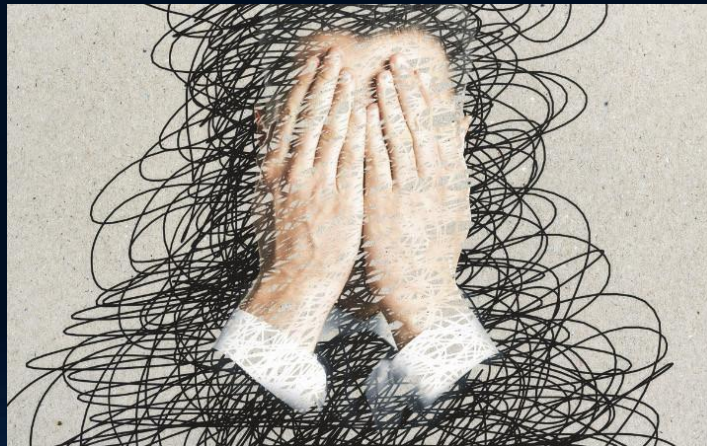
AI-Powered Digital Wellness Assistant

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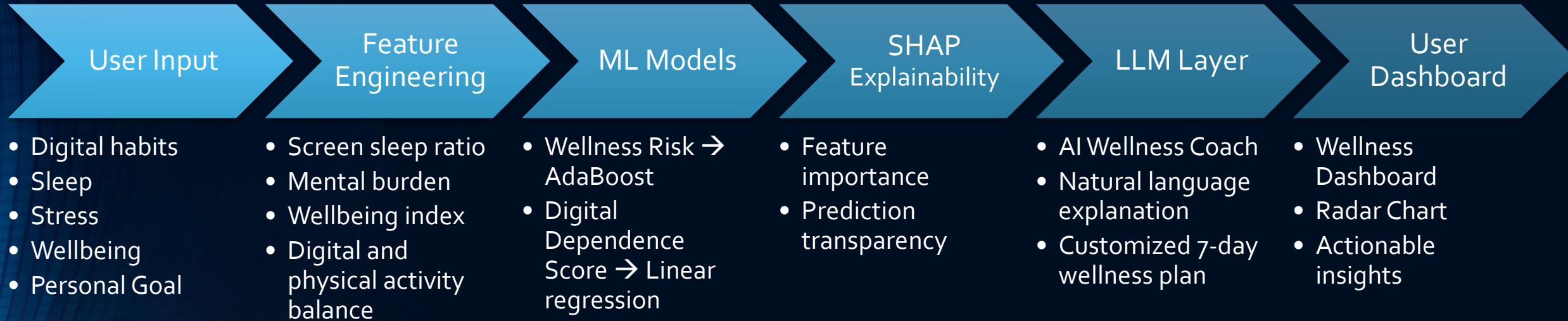
Machine Learning • Feature Engineering • LLM • Generative AI

How does modern lifestyle affect our wellness and our digital dependence?

- Excessive screen time
- Poor sleep habits
- Stress and emotional fatigue
- Lack of personalized wellness insights



Project Architecture



AI Digital Wellness Assistant

This wellness assistant estimates:

- Digital Dependence Score
- Wellness Risk Level

based on your digital habits, lifestyle, and wellbeing indicators.

About You

Step 1 of 4: About You

What should I call you?

Data Scientist

Hello Data Scientist!

Tell me something about yourself!

Gender

☒ Male ☐ Female ☐ Non-binary / Other ☐ Prefer not to say

Age

30

Where are you based?

☐ Africa ☐ Asia ☒ Europe ☐ Middle East ☐ North America ☐ South America ☐ Oceania

Highest education level completed?

☐ High School ☐ Bachelor ☒ Master ☐ PhD

Dataset & Prediction Targets

Features



Digital Habits

- Device hours
- Phone unlocks
- Notifications
- Social media usage



Mental Wellbeing

- Stress
- Anxiety
- Focus
- Happiness



Sleep & Lifestyle

- Sleep duration
- Sleep quality
- Physical activity



Demographics

- Age
- Region
- Education
- Income

Targets



Digital Dependence Score

Measures reliance on digital devices and online activities.



Moderate

Higher scores indicate greater digital dependence.



High Risk Flag

Identifies individuals at high risk based on behavioral and wellbeing patterns.



HIGH RISK

High risk users may benefit from early intervention.



Productivity Score

Evaluates focus, efficiency and ability to complete tasks.



Good

Higher scores reflect better productivity levels.

Machine Learning Pipeline

Classification

- Logistic Regression
- K-Nearest Neighbours
- Random Forest
- XGBoost
- AdaBoost
- Gradient Boosting

Regression

- Linear Regression
- KNN Regressor
- Random Forest
- XGBoost
- AdaBoost
- Gradient Boosting

Hyperparameter Tuning

RandomizedSearchCV

- n_estimators
- min_samples_split
- learning_rate
- colsample_bytree
- max_depth
- max_features

MODEL SELECTION & TARGET FINALIZATION

Choosing the right models for reliable predictions and focusing on the targets that matter most



1. WHY THESE MODELS?



AdaBoost Classifier (For Wellness Risk Prediction)

- ✓ Handles complex patterns in behavioural data
- ✓ Robust to noisy features and outliers
- ✓ Strong classification performance

★ Best balance of accuracy, recall and robustness for high risk classification.

Performance Summary

Metric	Score
Accuracy	0.883
Precision	0.820
Recall	0.540
F1 Score	0.650
ROC-AUC	0.782



Linear Regression (For Digital Dependence Score)

- ✓ Well-suited for continuous score prediction
- ✓ High explanatory power in the data
- ✓ Simple, interpretable and stable

★ Highest R² score indicates strong ability to explain variance in digital dependence.

Performance Summary

Metric	Score
MAE	0.708
RMSE	1.70
R ² Score	0.986

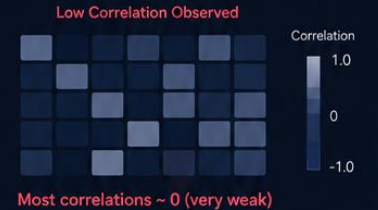


2. WHY PRODUCTIVITY SCORE WAS DROPPED



Low Correlation with Input Features

Productivity score showed very weak relationships with the available features, making it difficult to predict reliably.



Poor Model Performance (R²)



R² score is negative, meaning the model performs worse than predicting the mean.

✗ Not suitable for reliable predictions.



3. FINAL DECISION

Focus on the two most reliable and interpretable targets for maximum impact and user trust.



High Risk Flag (Wellness Risk Prediction)

Accurate classification of users at high risk for early intervention and support.



Digital Dependence Score (Continuous Score)

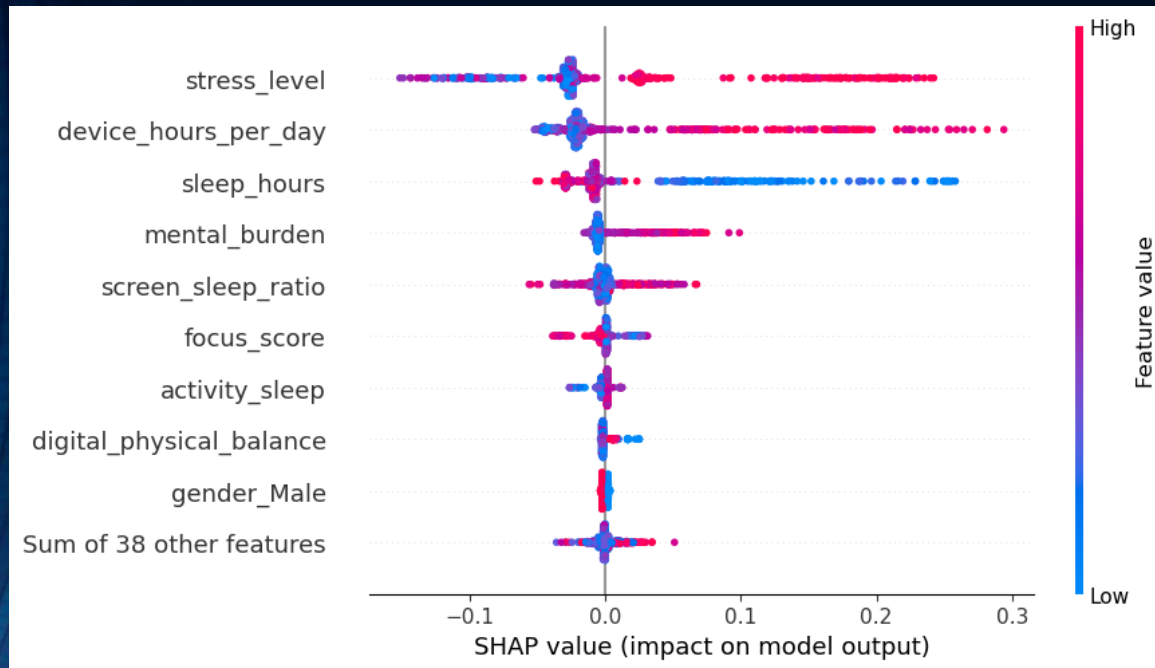
Strong predictive performance with high explanatory power and actionable insights.



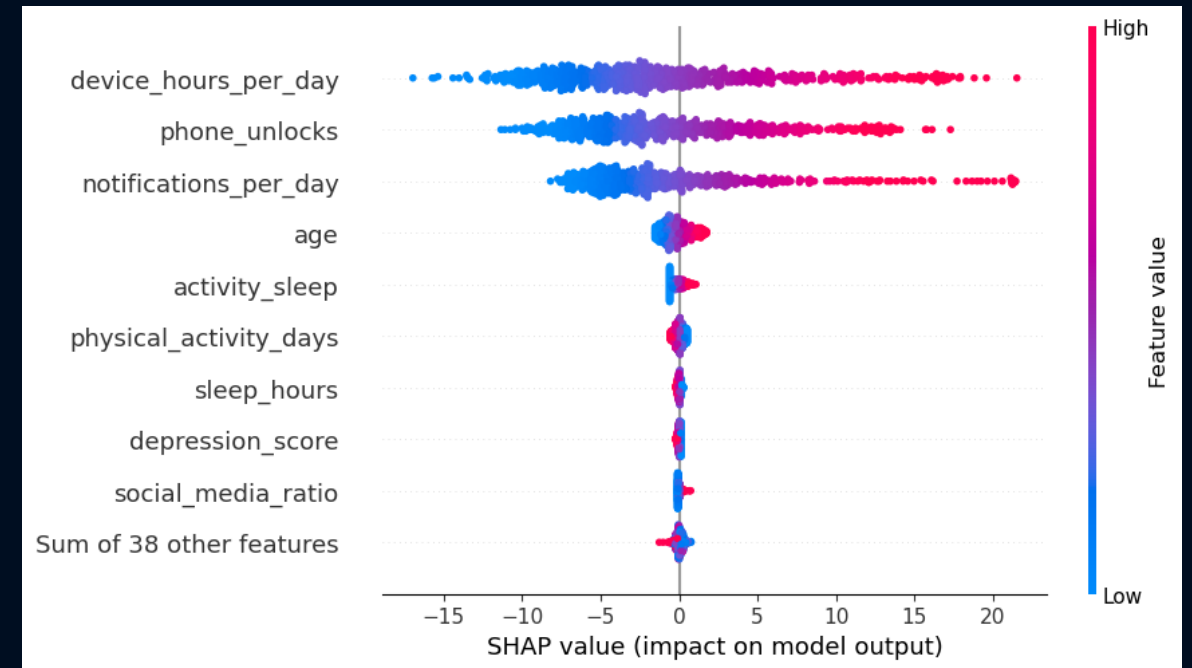
These two targets provide the most value, reliability and clarity for personalized wellness insights and recommendations.

SHAP

(High Risk Flag) Classification



(Digital Depedence Score) Regression



- SHAP (SHapley Additive exPlanations): explaining the output of a predictive model in Machine Learning, showing overall global feature importance
- It shows direction of feature influence and each dot represents one user from dataset training
- The top 5 SHAP features from each target were incorporated into the AI explanation and wellness coaching prompts.

LLM & Generative AI

AI WELLNESS COACH (LLM LAYER)

Turning predictions into personalized guidance and actionable plans

INPUTS TO LLM

-  **High Risk Probability**
Probability of high risk
-  **Digital Dependence Score**
Score of digital dependence
-  **Device Hours**
Hours on devices per day
-  **Physical Activity Days**
Days physically active
-  **Sleep Hours & Quality**
Sleep duration and quality
-  **Anxiety, Depression, Stress, Happiness, Focus Scores**
Mental & emotional indicators
-  **User Defined Goal**
Personal goal set by user

AI WELLNESS COACH (LLM FEEDBACK)

Analyzes all inputs and model outputs along with your personal goal to understand your overall wellness.

WHAT THE LLM PROVIDES

-  **Personalized Advice & Feedback**
Clear, user-friendly explanations tailored to your situation.
-  **Strengths & Areas to Improve**
Highlights what's going well and what to focus on.
-  **Recommendations**
Actionable, specific and easy-to-follow tips.
-  **7-Day Wellness Plan**
One tailored action for each day of the week.



WELLNESS PLAN (7 DAY PLAN)

The LLM creates one personalized action for each day of the week based on your inputs, risk, and goal.



User Input



AI Digital Wellness Assistant

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☐ 🇺🇻 South America
☐ 🌐 Oceania

Highest education level completed?

☐ 🎓 High School ☐ 🎓 Bachelor ☐ 🎓 Master ☒ 🎓 PhD

How would you describe your current financial situation?

☐ 🌱 Limited ☒ 😊 Comfortable ☐ 💰 Stable ☐ ✨ Very Comfortable

What best describes your daily routine?

☐ 🎓 Student
☐ 💼 Full-time Employee
☐ 🕒 Shift/Part-time
☐ 🏠 Caregiver/Home
☒ 🔍 Job Seeking



Digital Habits

Step 2 of 4: Digital Habits

What is your primary device?

☐ 📱 Android ☒ 🍏 iPhone ☐ 💻 Laptop ☐ 📺 Tablet

How much time do you spend on digital devices each day?

☐ 🟢 Very Little ☐ 🟡 Light Usage ☒ 🟠 Moderate Usage ☐ 🔴 Heavy Usage
☐ 🕒 Almost All Day

How often do you find yourself reaching for your phone?

☐ 🟢 Hardly Ever ☐ 🟡 Once in a While ☒ 🟠 Fairly Often ☐ 🔴 Very Often
☐ 🕒 Almost Constantly



Sleep & Lifestyle

Step 3 of 4: Sleep & Lifestyle

Your daily habits can strongly influence focus, stress, recovery, and overall wellbeing.

How active have you been recently?

☐ 🏃 Rarely Active ☐ 🚶 Lightly Active ☐ 🏃 Moderately Active ☒ 🏃 Very Active

How much sleep do you usually get?

☐ 😴 Very Little Sleep ☒ 🛌 Slightly Sleep Deprived ☐ 😊 Adequate Sleep
☐ 🌟 Well Rested

How would you rate your sleep quality recently?

☐ 😴 Very Poor ☐ 😞 Poor ☒ 😐 Average ☐ 😊 Good ☐ ✨ Excellent



Mental Wellbeing

Step 4 of 4: Mental Wellbeing

These questions help the assistant understand how you've been feeling emotionally and mentally lately.

Remember, there are no right or wrong answers, just choose what feels most accurate for you.

This is a safe space. You can be honest 😊.

How have your worries or anxious thoughts felt recently?

☐ 😌 Calm & Relaxed ☐ 😟 Occasionally Worried ☐ 😟 Somewhat Anxious
☒ 😟 Frequently Anxious ☐ 😟 Constantly Overwhelmed

How has your mood been recently?

☐ ✨ Positive & Motivated ☐ 😊 Mostly Okay ☒ 😞 Emotionally Tired
☐ 😞 Frequently Low ☐ 😞 Very Drained

How demanding or stressful has daily life felt lately?

☐ 🌱 Peaceful ☒ 😊 Manageable ☐ 😊 Busy but Coping ☐ 😟 Quite Stressful
☐ 🔥 Completely Overloaded

How satisfied and emotionally balanced have you felt recently?

☐ 😞 Very Unhappy ☒ 😞 Slightly Unhappy ☐ 😐 Neutral ☐ 😊 Generally Happy
☐ ✨ Very Happy

How focused and mentally clear have you felt recently?

☐ 😞 Unable to Focus ☐ 😞 Easily Distracted ☒ 😊 Average Focus ☐ 😊 Mostly Focused
☐ 🧠 Highly Focused



Personal Goal

What would you most like to improve?

I would like to worry less, be more cheerful, and be more productive.

Analyze

Result

Results

Nice work, Data Scientist.
Your current habits appear supportive of overall wellbeing. Consistency is key.

Wellness Dashboard

Wellness Risk

Risk Probability: 16.8%

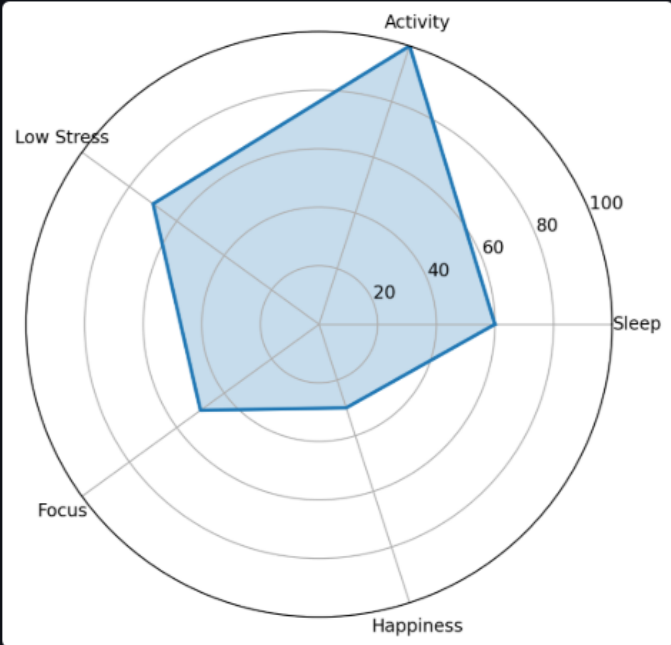
Low Risk

Digital Dependence

Dependence Score: 32.2

Low Digital Dependence

Wellness Profile



Your Strengths

Regular physical activity

Opportunities

Higher anxiety levels

AI Wellness Coach

Strengths:

- Maintains physical activity daily
- Moderate device usage (8 hours)

Areas to Improve:

- Sleep hours below recommended (6 hours)
- Low happiness and sleep quality scores
- Elevated anxiety and depression scores

Recommendation:

- Aim for 7-8 hours of quality sleep by establishing a consistent bedtime routine.
- Practice short mindfulness or relaxation exercises to reduce anxiety and boost mood.
- Take regular breaks during screen time to improve focus and productivity.
- Engage in enjoyable social or creative activities to increase cheerfulness.

Read Full Explanation

Summary: Your wellness indicators show a moderate level of stress and a solid, but improvable, sleep routine. Your device usage and frequent phone unlocks suggest high digital dependence, which may impact your cheerfulness and productivity.

Positive Habits:

- Low stress level (3/10) indicates you manage stress fairly well.
- Decent sleep duration of 6 hours each night supports basic rest.

Areas for Attention:

- High device usage (8 hours/day) and frequent phone unlocks could increase mental burden and worry.
- Screen-to-sleep ratio suggests screen time may be affecting your sleep quality and mood.

Next Step:

- Try setting specific phone-free times, especially before bedtime, to improve sleep and reduce mental clutter.

Your Personalized 7-Day Wellness Plan

Your Personalized 7-Day Wellness Plan

Goal Focus: This plan supports your goal by incorporating manageable daily habits that reduce worry, enhance mood, and boost productivity through better sleep, mindfulness, and structured work-rest balance.

Day 1:

- Practice a 5-minute morning mindfulness meditation focusing on your breath to reduce worry and start the day with calm awareness.

Day 2:

- Schedule 25-minute focused work sessions using the Pomodoro technique with 5-minute breaks to improve productivity and maintain mental clarity.

Day 3:

- Take a 20-minute outdoor walk after work to elevate mood and reduce anxiety while benefiting from physical activity.

Day 4:

- Establish a "digital sunset" by turning off screens 1 hour before bedtime to improve sleep quality and reduce digital dependence.

Day 5:

- Write down 3 things you're grateful for in the morning or evening to foster a more cheerful outlook.

Day 6:

- Add a 10-minute stretching routine in the morning to increase energy and reduce physical tension that can contribute to stress.

Day 7:

- Reflect on your week by journaling your accomplishments and areas for adjustment, setting positive intentions for the next week.

Additional Tips:

- Keep a consistent sleep schedule aiming for 7+ hours to improve mood and focus.
- Limit checking emails or social media to specific times to decrease digital overload.
- Prioritize hydration and balanced meals to support overall energy and brain function.

Export Results

Download Wellness Report

Key Takeaways & Future Improvements

- ✓ Combined ML + SHAP + LLM into one application
- ✓ Created personalized wellness insights
- ✓ Improved explainability and user experience
- ✓ Translated predictions into actionable guidance

- 🚀 Real-time, user-specific SHAP explanations
- 🚀 User progress tracking over time, history
- 🚀 Mobile app deployment
- 🚀 RAG-powered wellness recommendations

THANK YOU VERY MUCH FOR YOUR ATTENTION AND SUPPORT